



Republic of the Philippines
CAVITE STATE UNIVERSITY

CvSU Bacoor City Campus
City of Bacoor, Cavite
(046) 476-5029
www.cvsu.edu.ph



DEPARTMENT OF COMPUTER STUDIES

Thesis Panel Members

Department of Computer Studies
Cavite State University – Bacoor City Campus

SUBJECT: Request for Approval of Thesis Title Change

Dear Thesis Panel Members,

Good day,

We, the undersigned undergraduate researchers from Cavite State University – Bacoor City Campus, Department of Computer Studies, respectfully request the approval of our proposed thesis title change.

Our current thesis title is:

“FLOODSENSE: An Augmented Reality Mobile Application with Machine Learning-Based Flood Susceptibility Mapping for Bacoor City.”

Upon the recommendation of our Thesis Adviser and after careful consideration, we propose to revise it to:

“FLOODSENSE: An Android-Based Expert System for Flood Susceptibility Assessment and Pre-Event Preparedness in Bacoor City.”

The proposed revision reflects substantial changes to the system. The previously proposed **Augmented Reality (AR)** visualization and **Machine Learning (ML)**-based flood susceptibility classification have been removed. In their place, FloodSense will use a **Rule-Based Expert System** that applies flood-related knowledge, historical information, geographic data, and assessment rules provided or validated by authorized experts and institutional sources to determine the flood susceptibility of an area based on the user's selected rainfall intensity, rainfall duration, and location. The former Augmented Reality map will likewise be replaced by a dynamic and interactive color-coded map that allows users to move around the map, select areas, view evacuation centers, and observe changes in susceptibility levels according to the selected rainfall scenario.

The revised system also expands beyond flood susceptibility awareness by incorporating a **Decision Support System (DSS)** that provides appropriate pre-event preparedness guidance based on the Expert System result and the information or situation provided by the user. Features such as **Check My Area** and evacuation-center viewing will support location-specific preparedness, while a separate web-based administrative system will allow authorized administrators to maintain the Expert System knowledge and rules, Decision Support System content, geographic information, susceptibility zones, and evacuation-center records used by the Android application.

These changes shift FloodSense from an Augmented Reality and Machine Learning-based visualization tool into an expert-guided Android application focused on both flood susceptibility assessment and practical pre-event preparedness. Accordingly, the proposed title more accurately represents the revised objectives, functions, intelligent approach, and

overall scope of the study.

In this regard, we respectfully request the approval of the Thesis Panel Members for the proposed title change. We sincerely appreciate your continued guidance, recommendations, and consideration of our revised study.

Respectfully yours,

Reymart V. Goc-ong

Martin Lorenz D. Juanites

Romar T. Pulao

Researchers

Noted by:

EDAN BELGICA
Panel Member

09-02-26
Date

ELY ROSE PANGANIBAN – BRIONES, MIT
Panel Member

9/2/26
Date

Approved by:

CLARISSA V. ROSTROLLO
Defense Chairperson/Defense
Research Coordinator

9-2-26
Date

JOVELYN D. OCAMPO, MIT
Department Chairperson

9-7-26
Date